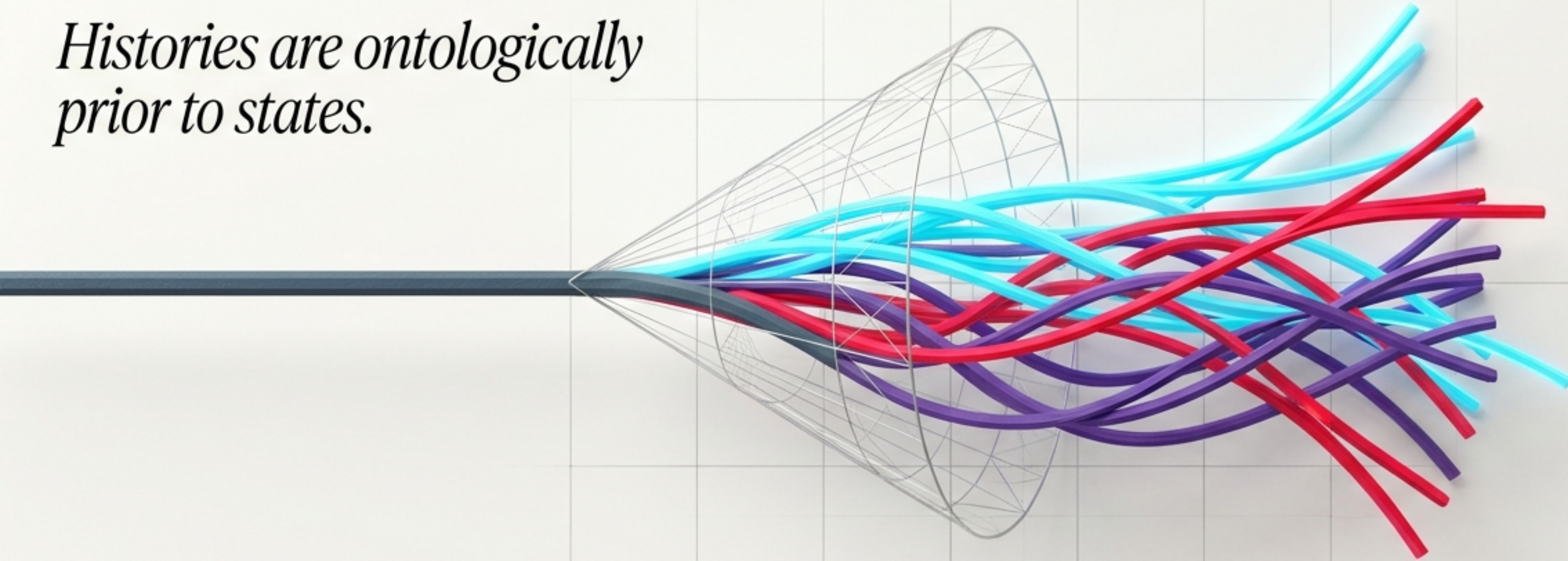


Spherepop: The Algebra of Reachable Computation

Histories are ontologically prior to states.



Primary takeaway:

Computation is not the manipulation of present values; it is the progressive restriction of future possibility.

Context:

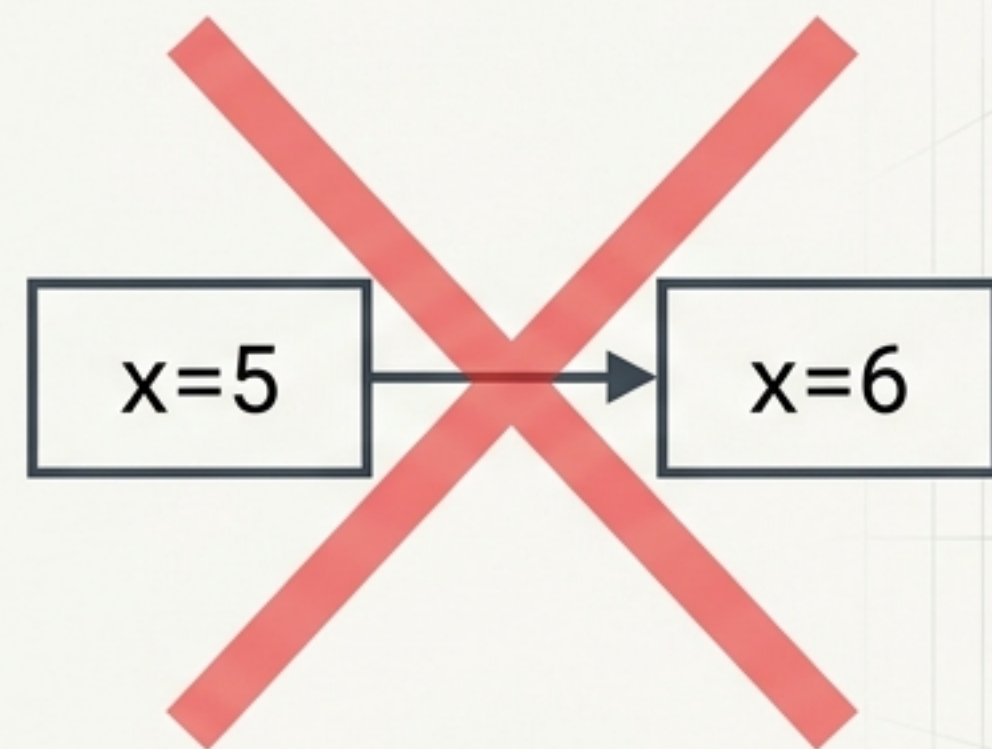
A note is an externalized continuation.

The Illusions of State and Memory



The Passive Theory of Notes

A grocery list does not record a past event;
it constructs a future action.



The State-Centric Assumption

A final value erases the sequence of
decisions that produced it.

Both paradigms make the same error: they mistake the terminal summary for the active substance. They value the snapshot while discarding the trajectory.

Histories Precede States



State is a derived notion—a collapsed residue of history.

A state is only instrumentally useful because it is a node in a navigable graph of continuations.

To prevent the loss of history, computation must treat the trajectory as its fundamental object.

A New Diagnostic for Computation

State-Centric Assumption

The Spherepop Inversion

Current State

Primary Object

Entire **Trajectory** (**History**)

Named locations holding values

Variables

Identifiers tied to historical
commitment events

Race conditions & Use-after-free

Bugs

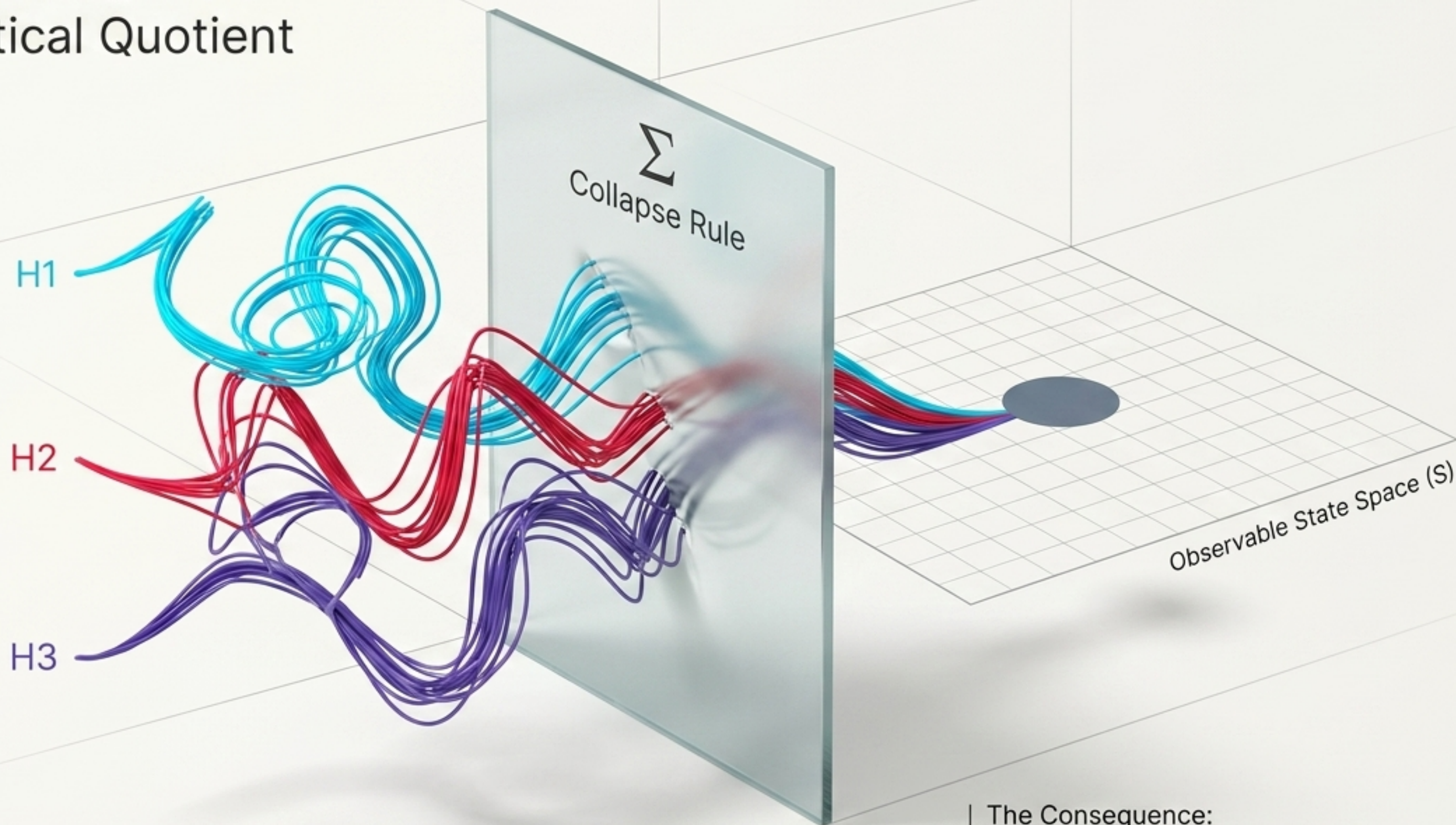
Admissibility certificate **failures**

Machine State manipulation

Compiler Target

Event Intermediate
Representation (IR)

The State Illusion is a Mathematical Quotient



The Proof:

For any non-trivial state space, multiple distinct histories map to the identical observable state.

The Consequence:

Any program exposing only the collapsed state to its user has silently discarded distinguishing information.

The Four Operators of Possibility Space

Pop
Commitment
Commits to a specific value,
permanently narrowing
the Option Space.



Refuse
Documented Inadmissibility
Excises a path from the
Admissibility set and permanently
embeds the reason.



Bind
Dependency
Creates a structural link
between two trajectory elements
without consuming either.



Collapse
Observation
Projects a history into an observable
state under a specific rule. Requires
an admissibility certificate.



The Conservation Law of Possibility



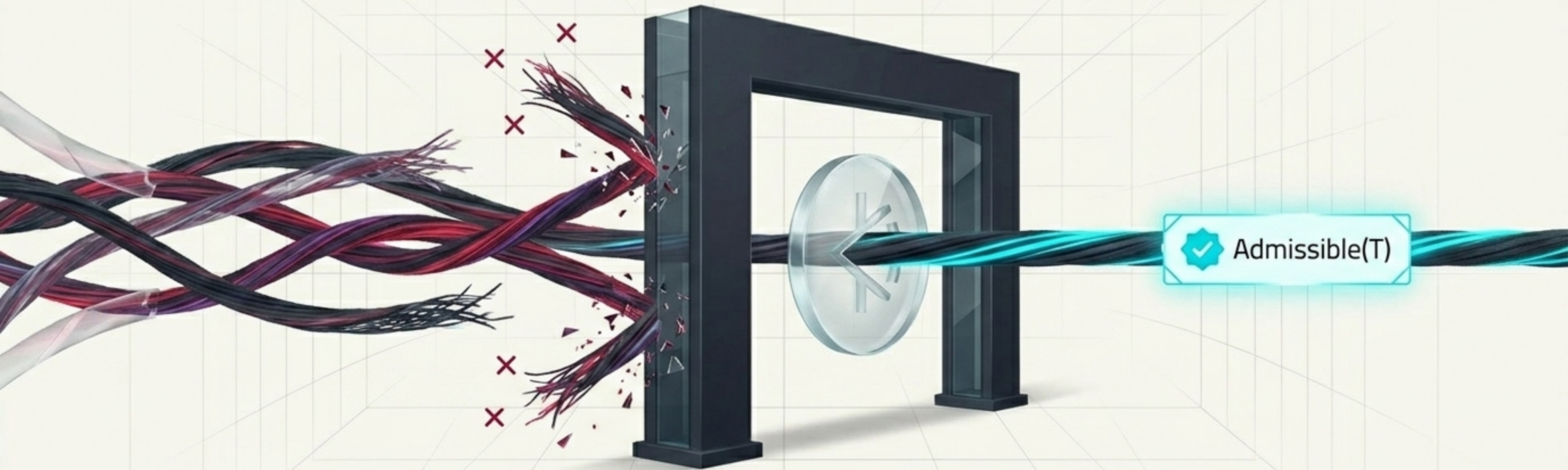
Possibility is never created or pop.

Possibility is never created or destroyed. It is either consumed by Pop or documented as inadmissible by Refuse.

Execution is strictly irreversible.

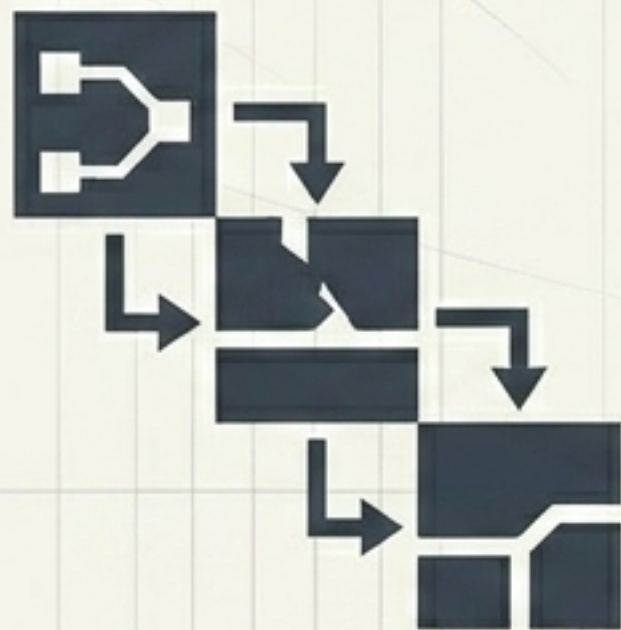
Execution is strictly irreversible. A world that could return to a prior state would undo history, which is not computation but its negation.

Types are Historical Proofs, Not Value Sets



- In a history-primary setting, a value is not merely a member of a set. It is the result of a path.
- A type is a mathematical proof (certificate) that the history producing the value did not violate any constraints.
- Implementation requires the Calculus of Constructions (CoC) so types can depend directly on the historical terms that produced them.

Rethinking the Computational Substrate



Compilers (Event IR)

Correctness is no longer just output equivalence. It requires Replay Equivalence—the compiled VM must construct the exact same history as the interpreter.



Garbage Collection

GC is not structural reference counting. It is admissibility-guided collapse, dropping dead history segments without breaking the type certificates of future continuations.



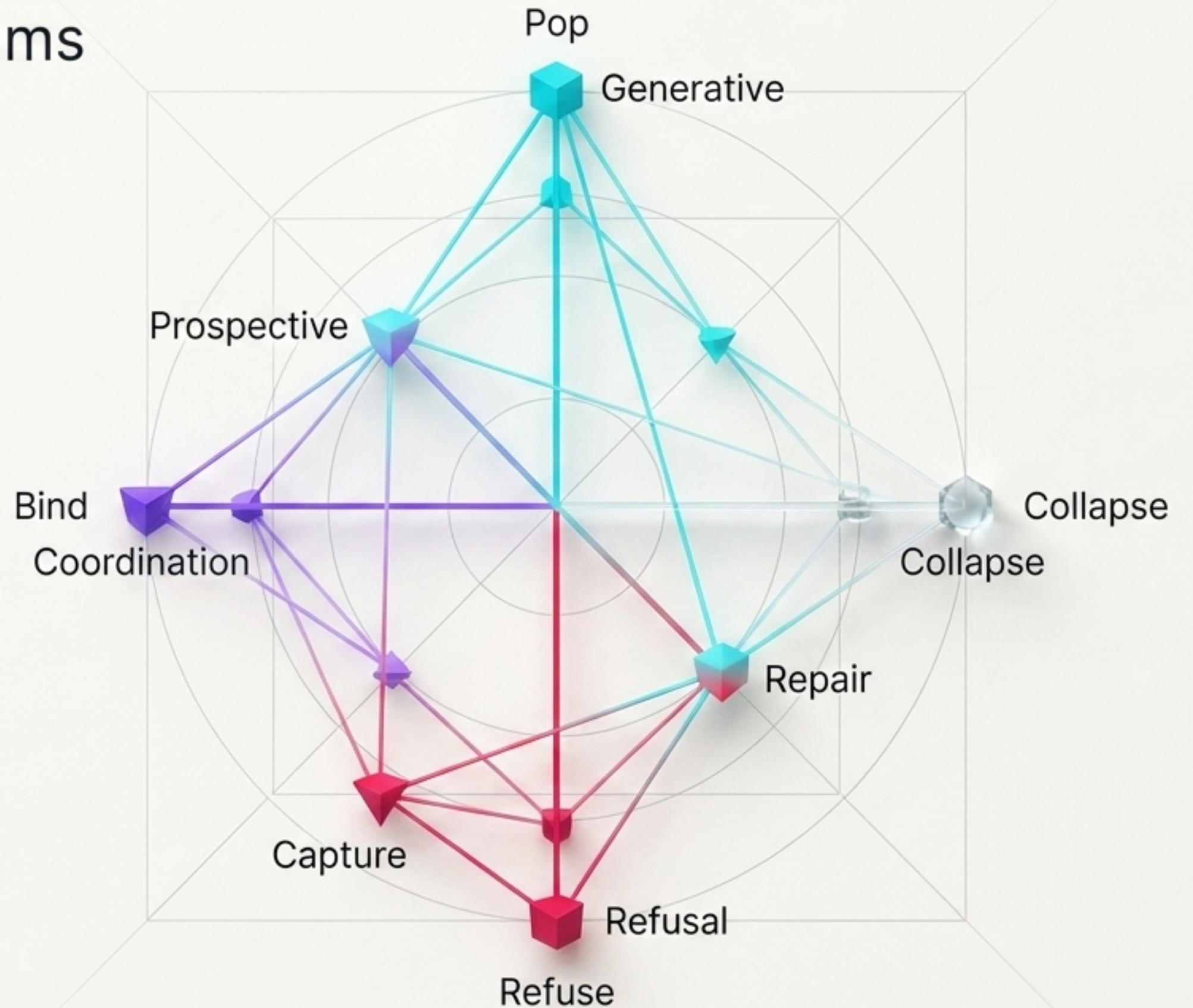
Error Correction

Errors are not thrown exceptions. They are Certified Refusals built into the history with redundant paths, allowing recovery without erasing the failure record.

Notes are Spherepop Programs

**Every note is a Spherepop program.
Every Spherepop program is a note.**

Human note-taking maps perfectly
onto the four mathematical operators
of possibility space.



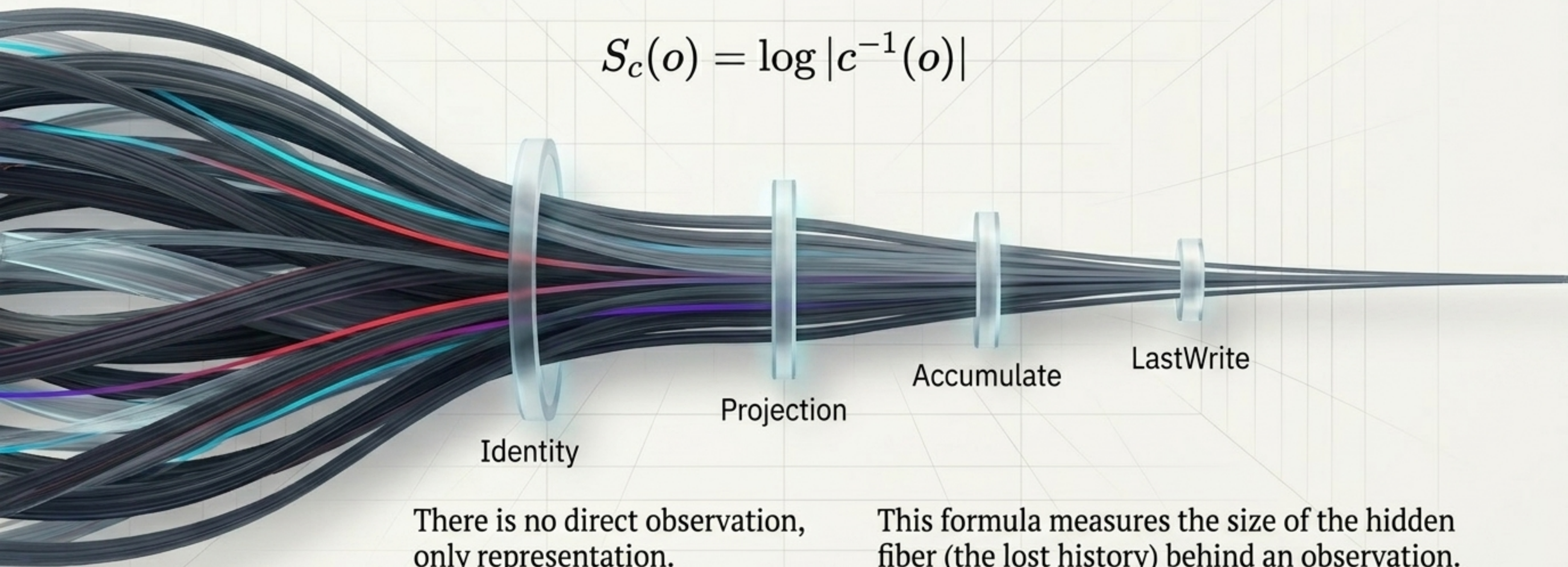
The Ephemerality Inversion

Y-Axis	Note Discarded	A discarded grocery list at the store exit. 100% Successful Note - Optimal
	Note Preserved	A perfectly preserved, unread manuscript. Unrealized Potential
		X-Axis
		Continuation Completed Continuation Unfinished

The Inversion: Ephemerality after completion is optimization, not failure. The celebrated surviving documents of antiquity survived precisely because their continuations were never completed.

Measuring the Cost of Observation

$$S_c(o) = \log |c^{-1}(o)|$$



Identity

Projection

Accumulate

LastWrite

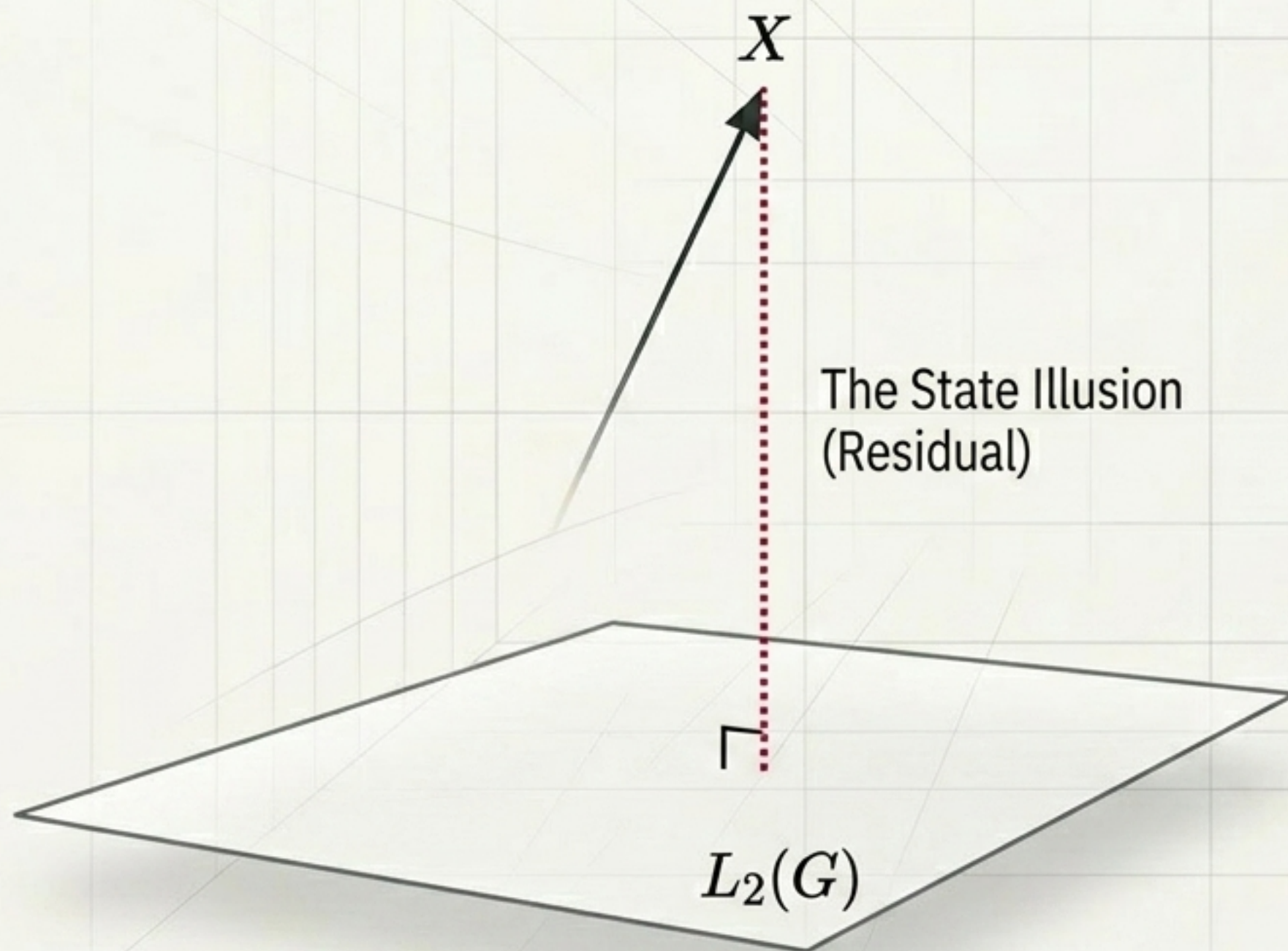
There is no direct observation,
only representation.

Finer rules preserve more history
but cost more to reconstruct.

This formula measures the size of the hidden
fiber (the lost history) behind an observation.

Notes act as right-adjoints, lowering this
reconstruction entropy.

Continuous Math for Discrete Logic



Discrete Logic
(Spherepop Collapse) is
mathematically
identical to

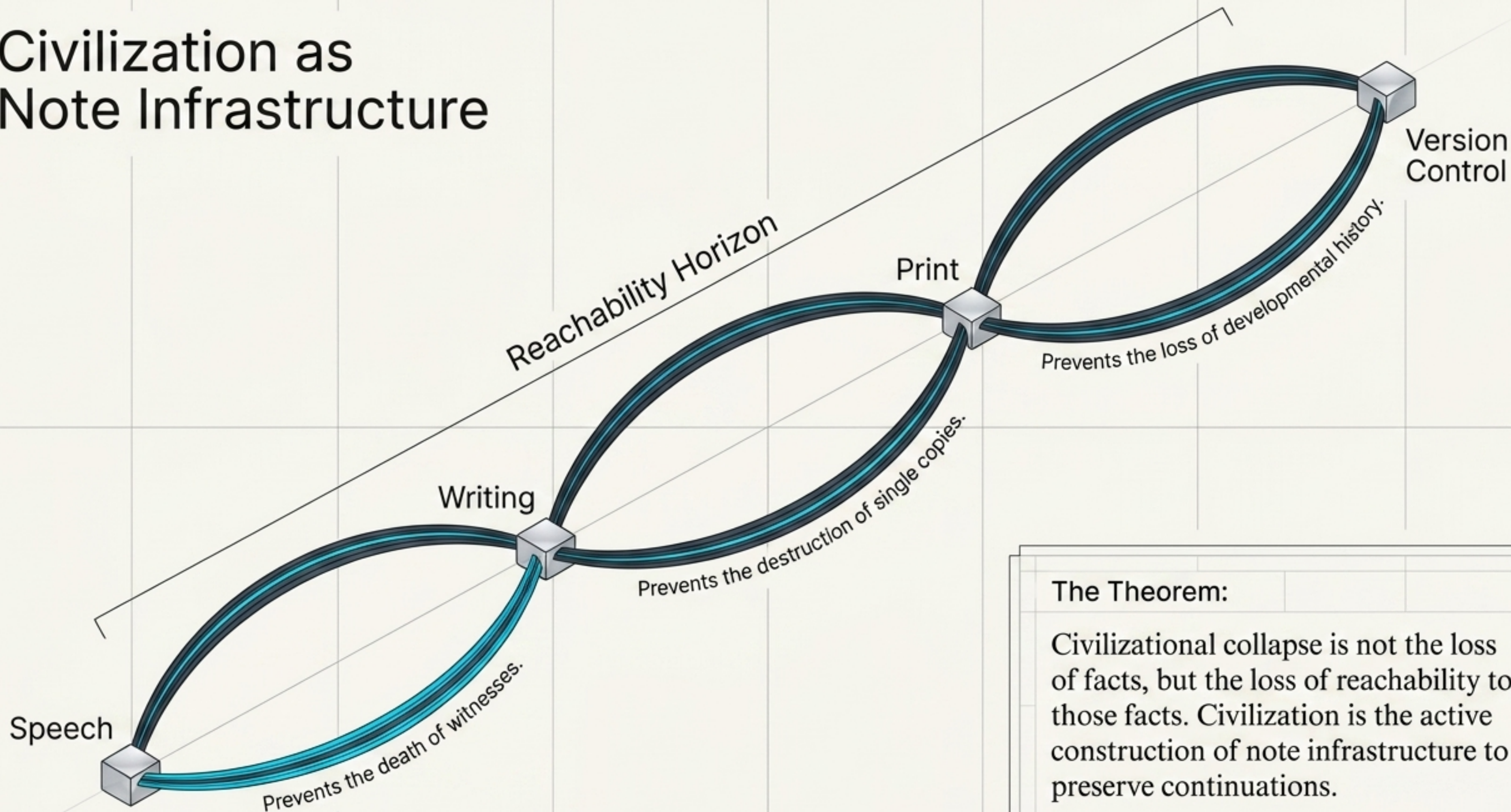
Continuous Math
(Kolmogorov
Conditional
Expectation).

Observational Entropy
(Discrete) maps perfectly
to

Variance
(Continuous).

The discarded history component is completely
orthogonal to the observational framework,
framework, making it mathematically invisible
to the observer.

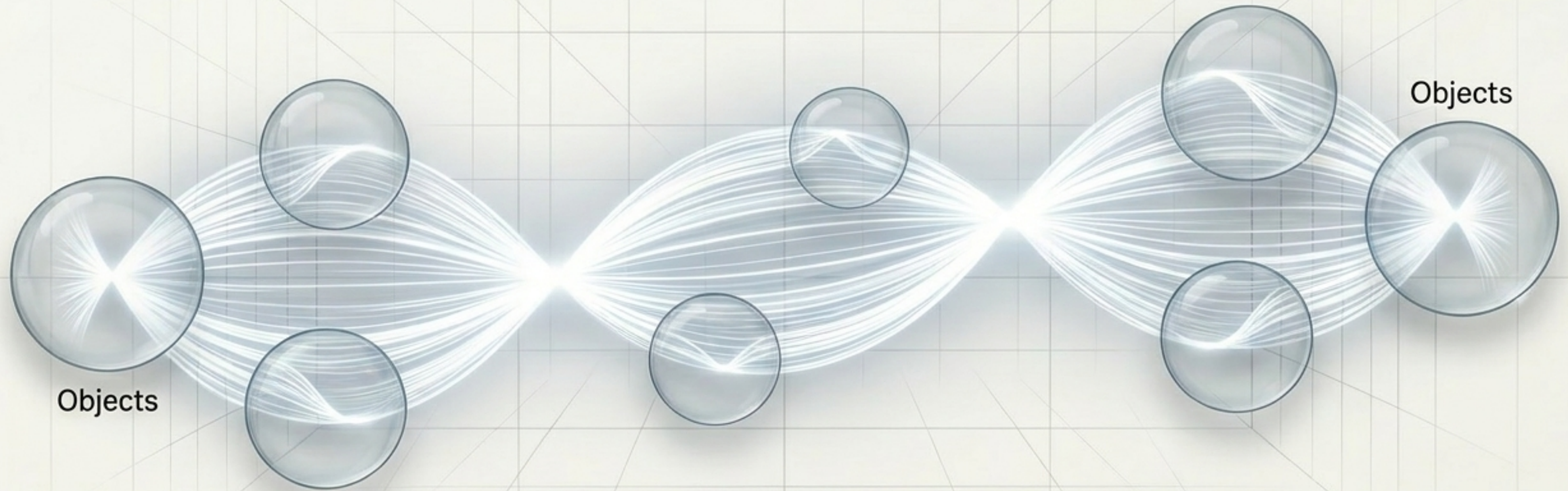
Civilization as Note Infrastructure



The Theorem:

Civilizational collapse is not the loss of facts, but the loss of reachability to those facts. Civilization is the active construction of note infrastructure to preserve continuations.

Reality is Reachable History



The Noun Fallacy: Objects do not fundamentally exist; they are merely the stable quotients of underlying histories. We mistake the noun for the process.

Reality is not a collection of primitive objects. Reality is a structured space of reachable histories.

Conclusion: Objects are what histories look like after sufficient collapse. Spherepop gives us the algebra to reverse it.